Under What Conditions Is Multi-AUV Coordination Feasible? A Constraint-Driven Review

LORENZO BACCHIANI, University of Bologna, Italy
ANDREA BEDEI, University of Bologna, Italy
ANDREA MELIS, University of Bologna, Italy
GIOVANNI PAU, University of Bologna, Italy
ROBERTO GIRAU, University of Bologna, Italy

Multi-AUV coordination underwater is bounded by acoustic communication, GPS-denied localisation, partial sensing, underactuated dynamics, finite energy and shared-channel contention, limits evaluations often leave implicit. This review characterises the resulting validation-deployment gap by deriving seven coupled physical constraints, defining the Information Contract imposed by coordination mechanisms, and coding 133 studies for evidence supporting compatibility claims. No reviewed primary study provides sufficient evidence to establish full-contract compatibility: each family leaves part of the underwater system unstated, and energy remains the residual gap. The findings yield six obligations for auditable coordination claims.

CCS Concepts: • **Computer systems organization** → *Fault-tolerant network topologies*; • **Networks** → *Peer-to-peer networks*; • **Hardware** → **Sound-based input / output**.

Additional Key Words and Phrases: Autonomous underwater vehicles, multi-AUV coordination, underwater acoustic communication, formation control, consensus, swarm robotics, task allocation, systematic review, feasibility analysis

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1 Introduction

Autonomous underwater vehicles (AUVs) have become central tools for oceanographic survey, environmental monitoring, infrastructure inspection, search and rescue and naval operations [49, 112, 121]. Many of these missions exceed the capability of a single platform, and multi-AUV systems promise meaningful operational gains: parallel task execution, redundancy, adaptive coverage and collective response to evolving mission conditions [27, 117, 170]. A substantial literature has grown around realising these gains, producing coordination mechanisms spanning formation control, consensus, swarm methods and task allocation. The question this paper asks is not what has been proposed, but what is physically admissible: which of these mechanisms can actually function underwater, and under what declared conditions?

Authors' Contact Information: Lorenzo Bacchiani, University of Bologna, Bologna, Italy, lorenzo.bacchiani2@unibo.it; Andrea Bedei, University of Bologna, Bologna, Italy, andrea.bedei3@unibo.it; Andrea Melis, University of Bologna, Bologna, Italy, a.melis@unibo.it; Giovanni Pau, University of Bologna, Bologna, Italy, giovanni.pau@unibo.it; Roberto Girau, University of Bologna, Bologna, Italy, roberto.girau@unibo.it.

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The answer is not obvious, because underwater coordination is not terrestrial or aerial multi-robot coordination transposed to a different medium. Submerged vehicles cannot rely on GPS, radio or WiFi at operational ranges. Communication is dominated by acoustic links offering bandwidths one to three orders of magnitude below terrestrial wireless standards, one-way propagation delays reaching seconds at kilometre ranges, and packet errors over intermittent, asymmetric connectivity [6, 43, 97]. Vehicles move in three dimensions under currents and hydrodynamic disturbances, without lateral thrust, on finite batteries, perceiving only a partial, uncertain local environment [49, 112]. These conditions impose hard physical ceilings on what coordination algorithms can achieve, regardless of their design.

These ceilings are rarely the primary subject of a coordination paper, so they tend to enter as background assumptions rather than declared scope conditions: connectivity is modelled as a fixed graph, vehicle state as ground truth, team sizes are kept small enough to avoid medium-access contention, and the infrastructure assets that make experiments viable, surface relays, acoustic beacons and synchronised clocks, are left implicit. These are reasonable simplifications for establishing algorithmic properties, but together they leave an uncharacterised distance between the conditions under which coordination mechanisms have been validated and the conditions the underwater environment imposes.

This paper characterises that distance. We derive *a structured model* of seven physical and operational constraints that jointly define the feasibility boundary any multi-AUV coordination mechanism must satisfy, grounding each in the acoustic channel, GPS denial, localisation uncertainty, vehicle dynamics, energy limits and partial observability. We define the *Information Contract* ($\mathcal{IC}$) each coordination family imposes on these constraints, and produce an *evidence taxonomy* distinguishing potential mismatches (where inferred requirements fall below the acoustic floor), conditional dependencies (satisfiable only under declared infrastructure), and scale-validation gaps (latent at small team sizes, emergent as fleet size grows).

Findings and Scope. Our analysis yields one structural finding and five family-specific ones, each qualified by the evidence in the reviewed corpus and by the coding rules documented in Section 4. The structural finding is that the corpus holds the pieces of feasible underwater coordination but does not assemble them into a full-contract demonstration: 0% of primary studies provide sufficient evidence to establish full-contract compatibility across all relevant contract and envelope dimensions, and the reason is that each family specifies its coordination mechanism while leaving one layer of the underwater system it runs on to assumption.

The five families leave different layers open. *Formation control* specifies its constraints one at a time but rarely together: read separately its studies declare communication in 65% and actuation in 72% of cases, but only 9% close the coupled communicate, keep fresh, localise and execute chain within a single study, and that closure collapses at localisation. *Consensus* proves agreement over a communication graph while declaring the observability and energy needed to realise that graph in 0% of its studies, a mathematically mature literature written over an un-instantiated substrate. *Swarm and bio-inspired* methods invert the usual trade, becoming the corpus minimum on the acoustic channel and the maximum on local sensing, but they ground that sensing in a real sensor in only 47% of their studies, replacing it elsewhere with virtual fields. *Task allocation* optimises the assignment and its cost while specifying vehicle localisation in 6% of studies, planning above an execution it assumes. *Architecture and support systems* is the supply side: it most consistently declares and fields the communication and localisation infrastructure the other families assume, holds the clearest conditionally compatible evidence and the most field validation, and comes closest to closing the contract.

The common residual is energy. It is the constraint most often left indeterminate, the one that in-water validation barely improves, and the one that even the architecture family, which contains demonstrated evidence on six of the seven constraints, does not demonstrate. These findings should be read as evidence about the declarations and validation conditions present in the literature, not as proof that existing algorithms are physically impossible; most gaps follow from reasonable scope simplifications made to establish algorithmic properties. From this characterisation we derive an actionable agenda: six obligations (Section 7) that make a multi-AUV coordination claim auditable by declaring the operating envelope and the information contract alongside the coordination law.

The remainder of the paper is organised as follows. Section 2 positions the review against prior surveys. Section 3 introduces the coordination paradigms and acoustic communication constraints needed to interpret the reviewed literature. Section 4 describes the review design, search strategy and screening criteria. Section 5 derives the constraint model. Section 6 compares coordination families under those constraints. Section 7 draws implications for future research. Section 8 analyses threats to validity. Section 9 concludes.

2 Related Work

Multi-AUV coordination has been surveyed repeatedly, but along two axes that the present work joins rather than repeats. The first organises the field by *coordination mechanism*, cataloguing the methods within a family and their open problems. The second reviews a single *enabling constraint*, such as navigation or the acoustic channel, in technical depth. Neither integrates the constraints into a per-study assessment of whether a coordination claim is physically admissible under its declared conditions. Table 1 positions this review against the closest works on both axes.

On the first axis, the broadest survey organises underwater multi-robot cooperation through a taxonomy of task-, motion- and measurement-space [172]; it is the closest in spirit to our aim and our source for the independent corroboration that the literature is simulation-dominated, but it classifies by the functional space in which cooperation emerges rather than by the information requirements coordination imposes on the channel. Within families the field is well reviewed: formation control, path following and path planning [49, 51, 146]; task allocation and multi-target search [117, 121]; finite-time and distributed coordination control [18]; and swarm and bio-inspired methods, from algorithm-and-hardware surveys [27] to cooperative-AI and bio-inspired treatments [14, 170]. The most recent swarm review classifies underwater approaches along four design dimensions, communication dependency, environmental adaptability, energy efficiency and scalability [92]. These reviews map what has been proposed and compare methods by design attributes; even the dimension-based classification orders approaches by their reported qualities rather than by whether a given study's declared envelope makes its coordination claim feasible.

On the second axis, individual constraints are reviewed in depth as enabling technologies. AUV navigation and localisation, the constraint we label C3, is surveyed from unbounded dead-reckoning drift through acoustic beacons, cooperative localisation and SLAM [85]; the acoustic channel and its medium access, the constraints we label C1 and C7, are covered by underwater acoustic sensor-network surveys of signalling, MAC and routing [101]. These works characterise one enabling constraint rigorously, but treat coordination as an application layer above it rather than as the object assessed against the constraint.

Closest in concern is Martorell-Torres et al. [80], which treats communication constraints as first-class; it is a primary architecture contribution, so the constraints serve as design inputs to one system rather than as a screen applied across a corpus. Collectively, prior work provides thorough taxonomies of coordination mechanisms and rigorous reviews of individual enabling constraints. Where these organise *what has been proposed* or characterise *one enabling technology*, this review organises *under what physical conditions those proposals remain admissible*, integrating the seven constraints into a per-study, cross-family feasibility assessment.

Table 1. Positioning against the most closely related surveys, along the two axes prior work follows: organisation by coordination mechanism (rows 1–5) and review of a single enabling constraint (rows 6–7); row 8 adds the closest communication-constrained system design. This work instead assesses individual studies against all seven constraints.

Ref.	Organising principle	Unit of analysis	Analytical contribution	Output / guidance
[172]	Functional-space taxonomy	Coordination family	Taxonomy of cooperation mechanisms	Open problems
[14, 27, 92, 170]	Swarm and bio-inspired mechanisms	Swarm family	Algorithms, hardware, and dimension-based classification	Open problems
[117, 121]	Task allocation and search	Coordination family	Taxonomy of allocation and search methods	Open problems
[49, 51, 146]	Formation and path control	Coordination family	Taxonomy of formation and path methods	Open problems
[18]	Distributed coordination control	Coordination family	Review of finite-time and distributed coordination-control methods	Open problems
[85]	Navigation and localisation (C3)	Enabling technology	Review of navigation and localisation techniques	Open problems
[101]	Acoustic sensor networks (C1, C7)	Enabling technology	Survey of signalling, MAC and routing	Open problems
[80]	Communication-constrained system design	Single system	One architecture under communication constraints	System architecture
This work	**Physical feasibility**	**Individual study**	**Information Contract; constraint model; evidence taxonomy**	**Reporting obligations**

3 Background

This section establishes the conceptual vocabulary used throughout the review. Section 3.1 introduces the main coordination paradigms found in the multi-robot literature. Section 3.2 describes the properties of underwater acoustic communication that make those contracts difficult to satisfy in submerged deployments.

3.1 Coordination Paradigms in Multi-Robot Systems

Multi-robot coordination concerns how local decisions, communication, sensing and action combine into useful collective behaviour [27, 49, 117, 121]. The literature contains four recurring paradigms, distinguished primarily by the information they require and the timescale on which they operate. A fifth, cross-cutting class, architecture and support systems, supplies the infrastructure the others presuppose and is treated alongside them. **Formation and leader-follower control** methods organise agents around a desired spatial structure, reference trajectory or leader-follower relation, through control laws that regulate relative position, velocity, heading or path-following error [5, 9, 17, 30, 32, 37, 44, 45, 49, 51, 93, 96, 120, 123, 126, 143]. They provide explicit geometric structure and predictable spatial relationships, at the cost of frequent exchange of leader or neighbour state. **Consensus and distributed control** methods define update rules through which agents progressively align local variables (heading, speed, position) with those of their neighbours [18, 19, 21, 29, 46, 58, 81, 106, 140, 144, 145, 163]. They support distributed decision-making without a central coordinator, but their feasibility rests on assumptions about graph connectivity, update timing and robustness to delayed or missing information. **Swarm and bio-inspired** methods emphasise local rules, simple individual behaviours and emergent organisation rather than precise global structure, coordinating through local attraction and repulsion, stigmergy, indirect environmental cues, minimal signalling or role differentiation [14, 27, 31, 59, 62, 102, 109, 113, 127, 136, 141, 159, 170]. They operate with sparse information exchange and tolerate partial observability, but offer weaker guarantees on convergence, optimality and task-completion time. **Task allocation and mission planning** methods operate at a higher decision layer: rather than continuously regulating motion, they assign vehicles to tasks, regions, targets or mission phases through auctions, market-based mechanisms, contract-net protocols, optimisation or learning [10, 40, 41, 66, 67, 71, 73, 75, 78, 91, 99, 117, 121, 131, 134, 156, 164, 168, 173]. Since a bid or assignment need not be exchanged at the rate of a control signal, negotiation burden and message reliability are the principal design concerns. **Architecture and support systems** supply the infrastructure the other four paradigms presuppose: cooperative localisation, acoustic communication frameworks, relay architectures, surface support [6, 23, 36, 43, 50, 61, 79, 97, 98, 103, 133, 135]. Coordination algorithms routinely assume that agents can estimate their own state, observe neighbours and exchange messages; in practice those capabilities depend on navigation accuracy, channel availability, clock synchronisation and relay support whose provision must be declared as part of any feasibility claim.

These classes differ not only in the collective behaviour they produce but in the IC they impose on the underwater environment. Evaluating a coordination method therefore means specifying that contract precisely; Section 5 defines it and uses it as the basis for the comparative analysis.

3.2 Underwater Acoustic Communication

Acoustic links are the primary communication medium for submerged vehicles. Radio-frequency signals attenuate rapidly in seawater, confined to very short ranges or to frequencies too low for data exchange; optical links offer high rates but need precise alignment and reach only a few tens of metres in typical conditions. Acoustic propagation reaches kilometres to tens of kilometres, but imposes constraints with no direct analogue in terrestrial wireless networks [6, 27, 43, 97].

Available data rates depend on range, carrier frequency and conditions, ranging from ≈100 bps to a few kbps at multi-kilometre ranges and rising to tens of kbps only at short range with wideband systems, one to three orders of magnitude below terrestrial wireless standards [6, 27, 43]. Sound propagates at $c \approx 1500$ m/s, so one-way delay reaches 670 ms at 1 km and exceeds 3 s at 5 km [24, 105]. Multipath propagation, Doppler spread, shadow zones, ambient noise and time-varying geometry produce packet-error rates of 10 to 50 % and links that may be intermittent or asymmetric over mission timescales [6, 27, 43, 97, 105]. With GPS unavailable, AUVs estimate state from Inertial Navigation Systems (INS), Doppler velocity logs, pressure sensors, acoustic ranging or periodic surfacing. These estimates drift over time, and their quality depends partly on the acoustic channel: cooperative localisation needs range or angle measurements carried over the same links as coordination messages [43, 97, 133]. Acoustic transmission draws energy that competes with propulsion, sensing and computation [6, 8, 109]. These properties are necessary background but, as Section 5 establishes on physical and control-theoretic grounds, not the feasibility boundary itself: they describe the channel in isolation, whereas the binding constraints on coordination prove to include the vehicle's kinematic envelope, the sensing footprint, medium-access contention at scale, and the couplings through which channel degradation propagates into localisation and energy. The constraint model of Section 5 is therefore developed through the hybrid procedure described in Section 4, combining physical constraints, networked-control requirements and recurrent assumptions observed in the literature.

4 Review Design and Search Strategy

We conducted the review as a reproducible secondary study: a query-based screening workflow identified peer-reviewed studies making a substantive contribution to coordination among multiple AUVs, unmanned underwater vehicles (UUVs) or underwater robotic vehicles, and a constraint-driven coding workflow compared the retained studies under a common extraction form. The protocol addresses two research questions:

- **RQ1.** What constraints fundamentally characterise underwater multi-AUV systems?
- **RQ2.** What evidence supports the compatibility of multi-AUV coordination mechanisms with the identified underwater constraints?

The constraint codebook was fixed before detailed family-level coding. All authors derived candidate dimensions from underwater physics and networked-control theory, checked them against review papers, enabling-technology literature and a pilot reading of primary studies, and froze the seven dimensions and decision rules in the extraction form before coding the 133 primary studies. Records were identified through structured searches in OpenAlex, Semantic Scholar, Scopus, CrossRef, IEEE Xplore and Google Scholar. Query strings combined underwater- and AUV-specific terms with the coordination paradigms of Section 3; the final search update was executed in May 2026, and only peer-reviewed records published from 2000 to 2026 and available in English were considered. We retained work whose primary contribution concerned multi-AUV, multi-UUV or underwater multi-robot coordination, including formation control, consensus, swarm methods, task allocation, mission planning and support architectures that supply coordination-enabling communication or localisation. We excluded single-vehicle planning or tracking, surface-vessel-only coordination without submerged acoustic constraints, terrestrial or aerial multi-robot work without underwater relevance, and adjacent networking, sensing or simulation papers that did not contribute to multi-vehicle coordination. Screening and coding were separated. Search results were normalised, deduplicated and screened first by title and abstract, then by full text; Figure 1 reports the resulting PRISMA flow. After deduplication, 489 records were screened at title and abstract level, 332 were excluded at that stage, 157 reports were assessed in full text, 11 were excluded after full-text assessment, and 146 met the corpus eligibility criteria: 133 primary studies subjected to detailed constraint-family extraction, and 13 review or survey papers retained for contextual synthesis rather than family coding. Hybrid papers were assigned by a dominant-primitive rule: the family is the coordination mechanism that carries the paper's main contribution and evaluation, while architecture/support-system labels are used only when the contribution is the communication, localisation or mission infrastructure itself rather than its incidental use. Every paper that entered family-level coding was coded independently by two coders on a common extraction form. Beyond bibliographic and family metadata, the form records, for each constraint, both the declared evidence and its coded status: for C1 the declared data rate, transmitted variable and its location in the text; for C2 the declared delay tolerance, the derived freshness requirement, the operational range and the acoustic propagation floor computed from it; for C3 the localisation method and drift rate; for C4 the vehicle actuation model and reference type; for C5 the sensing modality and declared range; for C6 the energy budget and power figures; and for C7 the validated fleet size and medium-access protocol. It also records the validation rung on an eight-level ladder, from analytical and idealised simulation through channel-aware simulation, hardware-in-the-loop, and pool, lake, sheltered-coastal and open-sea trials; the declared support infrastructure; and the study-level category with the aggregation rule that produced it. Before reconciliation, inter-coder agreement was high: Cohen's κ [26] was 0.990 for coordination-family assignment and ranged from 0.845 to 0.988 across C1–C7, with a mean κ of 0.948 over the eight coding variables and 97.9% of cells in exact agreement. The two coders then jointly re-analysed every paper against the primary full text and retained a value only when agreement

Table 2. Contents of the public data package [3]. The harmonised coding table is the authoritative source for every count and figure in the article.

Artefact	Records	Role
Screening master	489	Source and decision metadata for every screened record
Harmonised coding table	133	Per-study coding, authoritative for all counts
Review set	13	Survey papers retained for context, not detail-coded
Background references	27	References held out of the corpus denominator
Coder files	133	Independent pre-reconciliation coding from two coders
Scripts	–	Regenerate the inter-coder agreement tables and Section 6 figures

was reached. The harmonised table is therefore an adjudicated consensus record rather than a single-coder extraction, and although the study-level category is derived from C1–C7 by the documented aggregation rule, each final row is treated as a consensus decision, not an automatically generated value.

The complete package is public [3]; Table 2 summarises its contents. It provides the screening master, with database and query source, peer-review status, and title-abstract, full-text and exclusion decisions for every screened record; the harmonised coding table that is the authoritative source for all counts and figures; the two independent pre-reconciliation coder files; the uncoded review set; and the background references held out of the corpus denominator. It also ships the scripts that regenerate the inter-coder agreement tables and every Section 6 figure from the harmonised table, so that the article reports the synthesis while the repository carries the auditable extraction form, the per-paper decisions and the source-level evidence behind them.

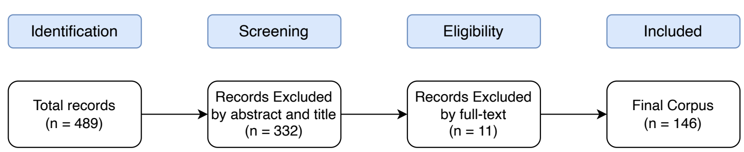

Fig. 1. PRISMA study selection flow diagram.

5 What constraints fundamentally characterise underwater multi-AUV systems?

Multi-AUV coordination imposes an $\mathcal{IC}$ on the underwater environment: the data a mechanism requires, the freshness at which it must arrive, the reliability with which it must be delivered and the sensing coverage that must underpin it. The central finding from the corpus is that the underwater environment places compound stress on these contracts, stress whose full scope no single study characterises, because none is positioned to address all seven constraints at once.

C1: Acoustic channel capacity. The data rates and packet-error statistics of Section 3.2 are the constraint every reader anticipates; what the corpus leaves largely unmodelled is their structural consequence: the communication graph is itself a time-varying, uncertain object, where an edge represents a possible acoustic path rather than a persistent channel [21, 46, 66, 171]. Methods that express neighbour exchange or bid propagation as inputs to a fixed connected graph therefore rely on channel assumptions whose feasibility must be declared. Architecture studies show that coordination is achievable when the acoustic pathway is designed explicitly, with content, rate, fallback and delivery assumptions [6, 41, 43]; studies that assume fixed graphs instead illustrate the scope condition C1 makes visible [37, 46].

Event-triggered designs acknowledge intermittency but still require bounded topology switches or eventual trigger delivery [44, 148, 150]. Swarm methods make C1 operative directly, matching payload and frequency to what the channel can realistically carry [109, 113].

C2: Information age. The propagation delays quantified in Section 3.2 set a hard floor, $\tau_{\min} = r/c$, on the information age any mechanism can achieve at operational range r: a message carrying leader position, neighbour velocity or a task bid may be operationally stale at reception even when delivery succeeds [44, 163]. $\tau_{\min}$ is the irreducible lower bound against which a mechanism's freshness requirement τ_{req} must be compared. Information age affects each paradigm differently: leader-follower formation is sensitive to leader-state age; consensus depends on the freshness of aligned variables; cooperative navigation requires time-aligned measurement association; task allocation depends on bid and observation currency [97, 111, 133]. Event-triggered designs reduce exchange frequency but do not remove the need to specify behaviour when an update is overdue, out-of-sequence or several delay periods old at consumption [44, 153, 171]. This is the underwater instance of a broader networked-control concern: delay, loss and sampling affect stability and performance, but the admissible freshness margin is mechanism-specific rather than a property of the network alone [54, 165, 167]. Physical experiments strengthen timing claims [124]; simulation evidence remains conditional on the assumed model [44, 163].

C3: Localisation and navigation uncertainty. AUV state estimates, produced by, e.g., INS, acoustic ranging or cooperative localisation, accumulate error at rates set by sensor quality, vehicle speed and acoustic-fix geometry [42, 85, 97]. Typical navigation errors range from 0.5 to 2% of distance travelled for DVL-aided vehicles, and errors as low as 0.1% can be obtained with large and expensive INS systems [4]; accordingly, low-cost INS units drift at roughly 0.1 to 1% of distance travelled, reaching tens of metres on multi-kilometre runs without acoustic correction. Thus, C3 is not a background sensor problem but a direct input to coordination feasibility.

The critical consequence is the feedback coupling with C1. Acoustic range measurements bound INS drift through cooperative localisation, so channel degradation (C1) propagates directly into localisation degradation (C3), which reduces the semantic validity of whatever the channel delivers. Cooperative localisation studies show this from the estimation side: delayed acoustic measurements and shared map updates must be associated with the correct past state before they can improve the present estimate [97, 133]. Early swarm-consensus experiments show that localisation uncertainty can be the binding performance constraint even for small teams with simple rules [58]. The infrastructural character of this constraint is visible in practice: cost typically precludes equipping every vehicle with high-precision inertial and Doppler sensing, so cooperative-navigation designs equip a small number of better-instrumented vehicles whose estimates anchor the rest [172], an asymmetry the architecture family later makes explicit (Section 6.5).

C4: Vehicle dynamics and underactuation. C4 constrains the set of responses a vehicle can physically execute: a transmitted command may arrive on time and at full accuracy yet still fail to produce the intended motion if it is kinematically inadmissible. Most operational AUVs cannot produce direct lateral thrust and are controlled through surge, pitch and yaw [49, 146]. Formation geometries trivially maintained by holonomic agents (fixed relative positions in three dimensions, instantaneous lateral displacement, station-keeping against transverse currents) may be physically unreachable by underactuated vehicles under hydrodynamic disturbances [5, 9, 17, 30]. The constraint operates at three levels: (i) lateral-flow resistance requires sway thrust the vehicle cannot produce [9, 30]; (ii) reconfiguration is bounded by turning radius and depth-control coupling [46, 162]; (iii) 3-D roll-pitch-yaw coupling makes formation maintenance qualitatively harder than the horizontal-plane case most reviewed studies address [5, 17, 56]. Studies that confront underactuation adopt path-following, line-of-sight guidance or cascaded-systems formulations feasible within the kinematic envelope [1, 9, 30, 45, 46, 52, 94, 95, 125, 162]. An information pathway that reliably delivers inadmissible references satisfies C1 and C2 yet still fails to enable coordination. The kinematic admissibility of the commanded reference is absent from coordination literature that inherits agent models from terrestrial or aerial domains.

C5: Partial observability and sensing range. Optical imaging is effective over 5 to 10 m in clear water and under 2 m in turbid coastal conditions [57]; forward-looking sonar resolves structure at tens of metres, but beam geometry and shadow zones leave portions of the near-field environment unobserved [86]. A vehicle therefore perceives only a partial, uncertain slice of the environment regardless of acoustic link status. C5 couples to C3 through a compound spatial resolution floor. A vehicle cannot contribute range measurements for cooperative localisation to a neighbour beyond its sensing range [15], so channel degradation and sensor range jointly bound the geometry over which cooperative positioning is feasible. The coupling also runs the other way: a vehicle accumulating INS drift reaches positional uncertainty without an acoustic fix, and when that uncertainty exceeds the spatial resolution of the task allocation scheme, the vehicle cannot reliably attribute a detection to a specific cell, so its bid becomes spatially ambiguous regardless of sensing range. Thus, C5 sets the *outer* boundary of what can enter, while C3 sets the *inner* boundary of what can be attributed with enough confidence to bid on. Neither bound is resolved by communication engineering alone, although team sensing, infrastructure, mission geometry or alternative sensor modalities may shift the effective observability envelope. Swarm and bio-inspired methods that use local attraction, repulsion, stigmergy or indirect cues [62, 141, 159, 170] are not merely communication-efficient: they are designed for agents that lack global state by construction, which makes them structurally aligned with C5. The CoCoRo architecture makes this explicit [100], and minimalistic one-bit alerts reinforce it, since such signals carry coordination value only when local sensing supplies the interpretive context [113]. For stigmergic methods the C3 coupling re-enters: a trace placed with positional uncertainty σ_1 and later read with accumulated drift σ_2 (which can reach tens of metres over multi-kilometre runs without acoustic correction [4]) produces, under independent errors, an RMS correspondence uncertainty of order $\sqrt{\sigma_1^2 + \sigma_2^2}$, which can exceed the interaction length assumed by the local coordination rule [141]. With correlated errors, the result also depends on covariance; for a relative error $e_2 - e_1$, the variance is $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, so common-mode bias may partly cancel while incremental or negatively correlated drift may enlarge the mismatch. The appropriate stochastic model is deployment-specific, but the coupling remains: stigmergic coherence depends on localisation, not only on the local rule.

C6: Energy budget. A detailed multi-AUV power budget places acoustic communication at roughly 2 to 10 W against propulsion loads of 50 to 100 W and sensing of 30 to 60 W [8]; recent AUV-class acoustic modems are likewise designed for sub-10 W operation [25]. Communication is thus usually a smaller slice of the budget than propulsion or sensing, but it is not free, and sustained acoustic exchange competes directly with the loads that dominate endurance: energy depletion before recovery is a primary operational risk on long-endurance missions [7]. Sparse-signalling, event-triggered and time-division multiple access (TDMA) designs reduce acoustic energy expenditure [109, 113, 153], but lower update rates directly worsen C1 and C2: fewer messages mean less information and greater staleness. Burlutskiy et al. make the coupling explicit, jointly optimising formation geometry, communication quality and power consumption and showing they are not independently tuneable [8]; energy-aware architectures with blockchain-based state management further expose how communication, energy and trust interact [70]. Specifying an update rate without declaring the energy budget over which it is sustainable yields an incomplete feasibility characterisation.

C7: Medium access and scalability. C7 characterises collective channel capacity as fleet size grows, a qualitative regime change driven by medium access control (MAC) contention and scheduling overhead, not simply C1 scaled by node count. The mechanism is physical, although its practical severity depends on fleet size, geometry and traffic regime: because a transmission's outcome cannot be sensed until it has propagated, the collision-avoidance window

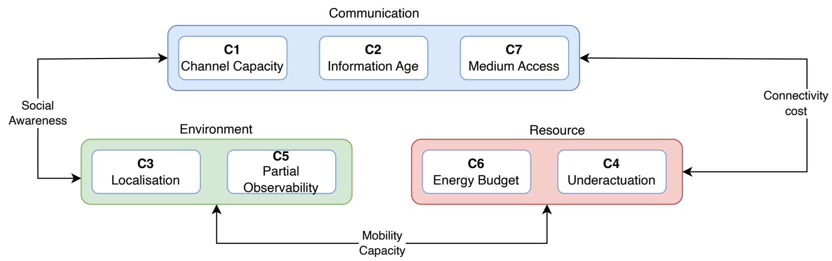

Fig. 2. The *Communication* group (C1, C2, C7) bounds *social awareness*: how much each vehicle can know about teammates (C1), how current that knowledge is (C2) and whether the shared medium sustains the exchange as fleet size grows (C7). The *Environment* group (C3, C5) captures localisation uncertainty and partial observability. The *Resource* group (C4, C6) captures kinematic and energy limits.

must scale with propagation delay rather than packet duration, so contention overhead grows with each added node in a way that has no terrestrial analogue [24, 87, 101]. Carrier-sense multiple access (CSMA) is particularly penalised in long-propagation regimes, where scheduling overhead can dominate useful exchange [110], an effect recent node-count simulations also expose [22]. Because acoustic capacity also decreases with distance, range and scalability cannot be separated cleanly at the system level [104]. The asymptotic consequence is formalised as an $O(1/n)$ per-node throughput bound for extended underwater acoustic networks, substantially worse than the $O(1/\sqrt{n})$ terrestrial-wireless result [76]; that bound is derived for large, geographically extended networks and does not transfer directly to the small teams in the corpus, but it supports the direction in which the mechanism above drives capacity under saturated, extended-network conditions. Most reviewed studies evaluate two to five vehicles, where this effect is modest beside the dominant C1 and C2 constraints, so protocols validated at this scale provide evidence for the tested regime, while larger-fleet claims require explicit MAC, traffic and geometry evidence. Rypkema et al. address this with a beacon-based architecture that decouples fleet size from acoustic channel load [97]; bio-inspired TDMA distributes slots adaptively [109] but validates large-swarm scalability mainly in simulation.

The seven constraints are not independent, and their couplings define the compound feasibility boundary any mechanism must satisfy. These couplings naturally aggregate the constraints into three functional groups (see Figure 2). No mechanism can therefore be assessed against a single constraint in isolation. The three groups act on distinct aspects of the underwater coordination problem, and their joint effect defines the feasibility boundary that any $\mathcal{IC}$ must respect.

5.1 The Information Contract Framework

The coupling analysis shows that coordination mechanisms are not defined only by their control law. They are also claims about an underwater system: that it can supply the information the mechanism consumes, keep that information fresh and reliable enough, and convert it into feasible motion. Let m denote the coordination mechanism under review. The claim made for m is meaningful only under a stated operating envelope,

$$\Omega = \langle r,\ n,\ G,\ L,\ M,\ K,\ H,\ I,\ \mathrm{Env}\rangle, \tag{1}$$

where r is the relevant communication or sensing edge length (or edge set, when links are asymmetric), n the fleet size, G the interaction topology and geometry, L the localisation architecture (e.g., INS/DVL, USBL/LBL, cooperative localisation), M the modem/channel/MAC assumptions, K the clock or synchronisation assumption (e.g., unsynchronised clocks, CSAC-backed OWTT, surface-disciplined time), H the mission horizon, I the declared infrastructure (e.g., surface relay, acoustic beacon, centralised mission manager), and Env the environmental regime (e.g., pool, lake/reservoir, sheltered coastal water, open sea). The edge length r, not the total mission scale, is what enters propagation delay and sensing reach; n, G and M shape contention and delivery probability; L, K and I determine what state can be known and under which support assumptions; and H bounds endurance claims.

Given m and Ω, the $\mathcal{IC}$ is the set of requirements that m imposes when evaluated in that envelope:

$$\mathcal{IC}(m \mid \Omega) \;=\; \left\langle\, \mathcal{D}_m(\Omega),\ \varepsilon_m(\Omega),\ \tau_m(\Omega),\ \Gamma_m(\Omega),\ O_m(\Omega),\ \mathcal{A}_m(\Omega) \,\right\rangle. \tag{2}$$

The arguments do not assert that every primary paper supplies closed-form functions of Ω. They make the scope explicit: each coded value is interpreted under the range, fleet size, topology, localisation, communication, infrastructure and environment assumed by the study. Each component captures one dimension of the contract:

- $\mathcal{D}_m$ is the *data object* required by the mechanism: e.g., relative position, neighbour velocity, task availability or a map update.
- ε_m is the *admissible uncertainty* of that object: the spatial, temporal or semantic tolerance within which $\mathcal{D}_m$ remains useful.
- τ_m is the *maximum acceptable information age*: the elapsed time beyond which a received, sensed or predicted value is no longer valid for the coordination decision.
- Γ_m is the *communication obligation*: payload size, message-generation rate, required recipients or fan-out, and the delivery probability needed within the τ_m window. Reliability is therefore not treated as a standalone p, because it is only meaningful for a declared load, topology and MAC/channel model.
- O_m is the *observability support* needed to instantiate $\mathcal{D}_m$ within ε_m: sensing footprint, modality, localisation accuracy and any infrastructure that supplies otherwise unavailable state.
- $\mathcal{A}_m$ is the *admissible and sustainable action set*: the subset of responses to $\mathcal{D}_m$ that the vehicle can physically execute and maintain over H, given its dynamics, actuation and energy limits.

The conditioning on Ω is therefore substantive, not decorative. The same mechanism may require the same nominal neighbour position $\mathcal{D}_m$, but its τ_m, Γ_m, O_m and $\mathcal{A}_m$ are evaluated differently in a 10 m USBL-supported pool formation and in a kilometre-scale open-sea formation. The constraint groups act as checks on different components: C1, C2 and C7 govern Γ_m and τ_m; C3 and C5 govern ε_m and O_m; C4 and C6 govern $\mathcal{A}_m$ over H. We use this framework as an evidence screen: for each paper, the coding asks whether the contract quantities and their envelope are declared, inferred, validated, infrastructure-dependent or absent. The evidence rung is recorded separately: it describes how directly the paper supports the claimed $\mathcal{IC}(m \mid \Omega)$, not a component of the operating envelope itself.

The framework is meant to be reusable. Given a new coordination mechanism, an author instantiates each contract component and then checks it against the governing constraints. Table 3 supports that step: it summarises, for each family, the forms the components typically take across the reviewed corpus, using the harmonised coded fields: the transmitted variable for $\mathcal{D}_m$ and Γ_m, the medium-access protocol and validated fleet size for C7, the derived freshness rule for τ_m, the localisation method and sensing modality for O_m/ε_m, and the actuation model and energy figures for $\mathcal{A}_m$. It describes what each component *is* in each family, not how completely it is specified; the adequacy of that specification is the subject of Section 6. A reader can read down a family column to see how the contract is normally realised, and across a row to see how one obligation shifts between coordination styles.

Table 3. Instantiating the $\mathcal{IC}$ by coordination family. Each cell gives the form a contract component (left, with its governing constraints) typically takes in that family. $\mathcal{D}_m$ is the data object; Γ_m the communication obligation (C1 channel, C7 access and scale); τ_m the freshness bound (C2); O_m/ε_m the observability support (C3 localisation, C5 sensing); $\mathcal{A}_m$ the admissible sustainable action (C4 actuation, C6 energy).

Component · constraint	**Formation**	**Consensus**	**Swarm / bio-inspired**	**Task allocation**	**Architecture / support**
$\mathcal{D}_m$	Leader/neighbour relative pose, velocity	Neighbour state vectors (position, velocity)	Local cues: sync pulses, relative position, stigmergic markers	Robot status, task bids/assignments	Infrastructure payload: timing, ranges, positions, status
Γ_m·C1	State broadcast at control rate	State packets over a fixed/switching graph	Short-range RF/optical or sensing-mediated	Periodic status/bid broadcast	Acoustic modems, scheduled compressed payload
Γ_m·C7	Broadcast to small simulated or tested teams	Small static/switching graph topology	Local broadcast/slotting; large-n mainly simulation	TDMA or negotiation schedule; mostly simulated teams	Named MAC/TDMA, few platforms
τ_m·C2	Sample/update period or declared delay bound	Bounded/sampled delay in convergence proof; often no real-time deadline	Event/interaction-paced updates	Allocation/replanning cycle	Declared message cadence
O_m/ε_m·C3	Model-based own-state estimation	Neighbour state obtained by exchange	Local relative position	Known, extrapolated or dead-reckoned position	USBL/DLBL, GPS-at-surface, DVL/INS
O_m/ε_m·C5	Assumed relative state or onboard sensing	State exchanged, not sensed	Onboard relative sensing	Mission-payload sonar	Ranging and platform sensors
$\mathcal{A}_m$·C4	Under/fully-actuated dynamics tracking a path	Reduced/integrator dynamics to a leader/target	Point-agent/kinematic motion	Constant-speed kinematic vehicle following optimiser output	Real-platform integration, control external
$\mathcal{A}_m$·C6	Control-input effort	Control-input effort	Battery-status awareness	Per-AUV energy as an allocation cost	Battery/endurance specs or comm-energy models when declared

5.2 From Contract to Evidence: Coding Categories

The $\mathcal{IC}$ framework makes explicit what a coordination mechanism demands, but a demand is not a proof of supply. We therefore classify the evidence for the system claim made by each study, not the intrinsic value of the algorithm in isolation. A mechanism may be feasible only under an envelope that includes, e.g., a surface relay, acoustic beacon, synchronised clocks, or other support infrastructure. When declared, these assets are part of Ω and define the scope of the claim; when implicit or absent, they leave an evidence gap about where the claim generalises.

Evidence categories. We assign each coded primary study to one of four mutually exclusive categories, all interpreted relative to the declared Ω; none denotes unconditional feasibility. *Demonstrated compatible* is the strongest category: the relevant $\mathcal{IC}$ components and operating envelope are declared, and the paper provides evidence representative of the claimed scope. Infrastructure dependence does not preclude this label: a relay-, beacon- or clock-supported system can be demonstrated compatible if the contract is validated with those assets in place. *Conditionally compatible* is used when declared infrastructure or scope restrictions plausibly supply an otherwise open contract dimension, but the complete contract is not directly validated at the claimed scope. This category is not a penalty; it records that feasibility depends on reproducible system conditions and should not be read as generalising outside them without re-evaluating the contract. *Indeterminate* means that the paper does not provide enough information to decide whether the relevant contract can be supplied. This is the conservative outcome for underreported timing, localisation, observability, energy or scale evidence. Missing values therefore remain missing evidence; they are not converted into physical incompatibilities. *Potentially* *incompatible* means that the available reported or defensibly inferred values conflict with at least one constraint under the stated envelope. Examples include a freshness requirement below the acoustic propagation floor at the relevant edge length, a communication load outside the declared channel model, or a requested reference outside the specified vehicle's admissible action set. The qualifier "potentially" is deliberate: the category identifies a conflict in the claim as reported, not a direct observation that the system failed in deployment.

Study-level decision rules. The coding first reconstructs m, Ω and the observable parts of $\mathcal{IC}(m \mid \Omega)$. The relevant distance for C2 is the communication or sensing edge length r, not the total mission scale. C1, C2 and C7 are assessed through Γ_m and τ_m: payload, message rate, recipients, channel/MAC model, fleet size and timing tolerance must be declared or inferable before an incompatibility is assigned. In particular, a controller sampling period is not by itself a maximum admissible communication age; absent a declared tolerance or a mechanism-specific derivation, the freshness evidence is indeterminate. C3 and C5 are assessed through ε_m and O_m: local sensing limits matter, but team sensing, cooperative mapping, relayed detections and infrastructure can conditionally supply state that a single vehicle cannot observe alone. C4 is assessed at study level by comparing the requested reference with the declared vehicle model; a family label alone is not evidence of inadmissible motion. C6 is assessed only when rate, mission horizon and power or energy assumptions are sufficiently specified.

Overall assignment follows a conservative rule. A study is demonstrated compatible only when all relevant contract components are declared and validated for its envelope, including any declared infrastructure. It is conditionally compatible when declared infrastructure or scope restrictions support, but do not directly validate, one or more otherwise open dimensions. It is potentially incompatible when reported or defensibly inferred quantities conflict with a constraint under the stated envelope. Otherwise it is indeterminate. The same rule is applied to every coordination family, so missing timing evidence in formation, instantaneous global state in learning-based simulation, and unreported observability in task allocation are all treated as evidence gaps unless the paper provides enough information for a stronger category.

6 What evidence supports the compatibility of multi-AUV coordination mechanisms with the identified underwater constraints?

Section 5 defined compatibility as a claim about a coordination mechanism together with the underwater system that must supply its information, keep that information usable, and execute the resulting actions. We therefore begin the analysis in the space of the Information Contract. For each primary study, we reconstruct the declared operating envelope Ω and the observable components of $\mathcal{IC}(m \mid \Omega)$, then ask which parts of that contract are demonstrated, conditionally supplied, left indeterminate, or in potential conflict with the stated envelope.

Figure 3 gives the main answer. The evidence gap is not a generic lack of validation spread uniformly across the corpus; it is concentrated in specific parts of the contract. The data object $\mathcal{D}_m$ is not shown as a separate row because it is usually identifiable from the coordination primitive itself: leader state, neighbour state, target observation, task bid, map update or environmental trace. The feasibility question is whether that object can be communicated as required (Γ_m), consumed while still fresh enough for the decision (τ_m), observed or localised with bounded uncertainty (O_m/ε_m), and converted into admissible sustainable action ($\mathcal{A}_m$). Indeterminacy dominates all four displayed components: 76% of studies are indeterminate for the communication obligation Γ_m, 54% for freshness τ_m, 93% for observability and uncertainty O_m/ε_m, and 76% for admissible and sustainable action $\mathcal{A}_m$. This is the central empirical finding. The corpus usually specifies what the coordination law computes, but much less often specifies the underwater system contract that makes the computed information available, current, observable and executable. The matrix also distinguishes missing

Information Contract component	Evidence category (papers %) DC	CC	IND	PI
Γ_m Communication obligation	6%	16%	76%	2%
τ_m Freshness	8%	37%	54%	2%
$\mathcal{O}_m/\varepsilon_m$ Observability/uncertainty	1%	6%	93%	0%
$\mathcal{A}_m$ Admissible/sustainable action	2%	7%	76%	15%

Fig. 3. Information-contract evidence matrix for the 133 primary studies. Rows are components of $\mathcal{IC}(m \mid \Omega)$; columns are component-level evidence categories, ordered from strongest to weakest support: Demonstrated Compatible (DC) > Conditionally Compatible (CC) > Indeterminate (IND) > Potentially Incompatible (PI). Component status is obtained from the limiting status of the constituent constraints: Γ_m is limited by the worst of C1 and C7, τ_m by C2, $\mathcal{O}_m/\varepsilon_m$ by the worst of C3 and C5, and $\mathcal{A}_m$ by the worst of C4 and C6. Cell values are percentages of primary studies. $\mathcal{D}_m$ is omitted because the data object is usually identifiable from the coordination primitive; the feasibility question is whether it can be communicated, kept fresh, observed with bounded uncertainty, and converted into admissible sustainable action.

evidence from potential conflict. Potential incompatibility appears only when reported or defensibly derived contract quantities conflict with a physical or operational constraint under the stated envelope. It is most visible in Γ_m, τ_m and $\mathcal{A}_m$, where communication load, information age, actuation assumptions or energy requirements can be compared against the declared envelope. By contrast, $\mathcal{O}_m/\varepsilon_m$ is almost entirely indeterminate rather than potentially incompatible: most papers do not provide enough sensing-range, localisation-error or observability evidence to test the component at all. Conditional and demonstrated component-level evidence does exist, especially for communication and freshness in studies that declare modems, schedules, relays, clocks, localisation systems or restricted operating conditions. The problem is therefore not that the corpus lacks useful evidence; it is that the evidence is usually partial at the contract level. This contract-level diagnosis explains the paper-level categories reported next. The empty Demonstrated Compatible category is not the main result by itself. It is the paper-level consequence of the profile in Figure 3: no individual study closes all relevant contract and envelope dimensions at once, although many close important subsets.

Figure 4 gives the paper-level closure result implied by the contract matrix. Across the 133 primary studies, 79% are indeterminate, 17% are potentially incompatible and 4% are conditionally compatible; 0% report enough evidence to establish Demonstrated Compatible status across all relevant contract and envelope dimensions. This zero is informative only in the full-contract sense: it shows that no individual paper makes the complete underwater feasibility claim auditable within a single declared scope. It does not mean that the corpus lacks useful algorithms, meaningful experiments or credible system components. On the contrary, the conditionally compatible cases show that compatibility evidence becomes strongest when the underwater support system is part of the claim. The distribution is not uniform. Potential incompatibility appears in 25% of formation studies, 30% of consensus studies and 11% of task-allocation studies; together, formation and consensus account for 91% of potentially incompatible cases. These are high-information-contract families: their claims often depend on frequent state exchange, graph realisation, timing assumptions or references that must be executed by underactuated underwater vehicles. Conditional compatibility follows the opposite pattern. It appears when a study declares the system boundary that supplies the needed capability, with three of the

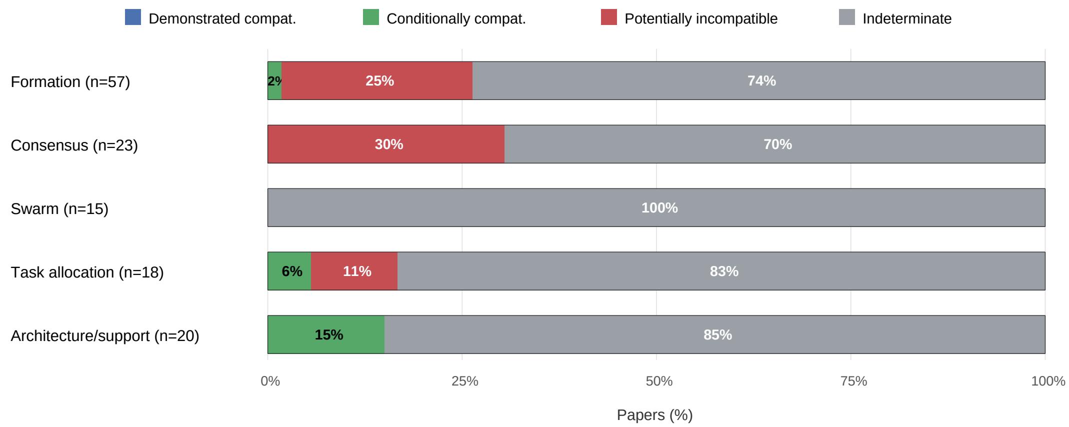

Fig. 4. Evidence category assigned to the 133 detailed-coded primary studies, grouped by coordination family. Percentages inside bars are family percentages. Demonstrated Compatible is a full-contract category: it requires all relevant $\mathcal{IC}$ and operating-envelope dimensions to be declared and supported for the claimed scope. No study reaches this category, but several provide conditional or dimension-specific evidence.

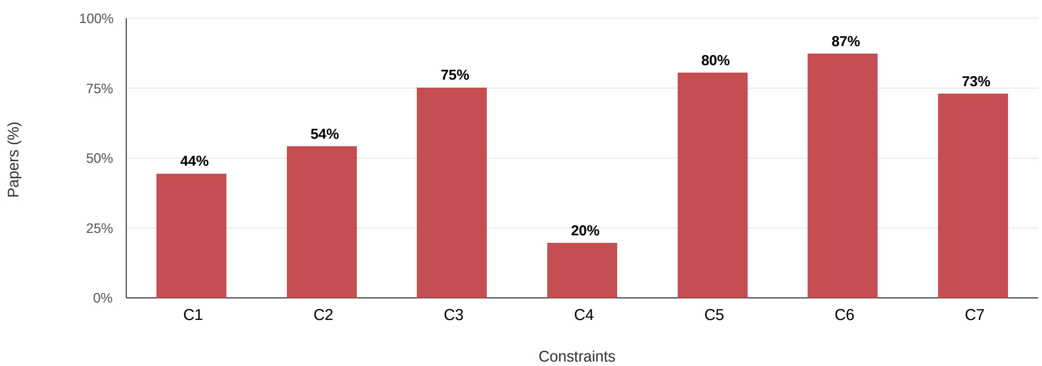

Fig. 5. Recurring missing or underreported evidence dimensions across the 133 primary studies. Bars show the percentage of studies coded indeterminate on each constraint in the harmonised evidence table.

five cases in architecture/support and one in formation after the digital-twin field study is credited for functionally supplying its timing contract within the tested stack. The main finding is therefore not that no paper receives the strongest label. It is that compatibility evidence is strongest where the underwater support system is declared, and remains unresolved where key elements of $\mathcal{IC}(m \mid \Omega)$ are absent.

Figure 5 explains which constraints drive the contract-level indeterminacy. The missing dimensions are not peripheral implementation details; they are the quantities that determine whether $\mathcal{IC}(m \mid \Omega)$ can be instantiated underwater. Energy evidence is missing or indeterminate in 87% of studies, sensing range or equivalent observability support in 80%, localisation/drift evidence in 75% and MAC/scale evidence in 73%. C4 is the relative exception: only 20% are

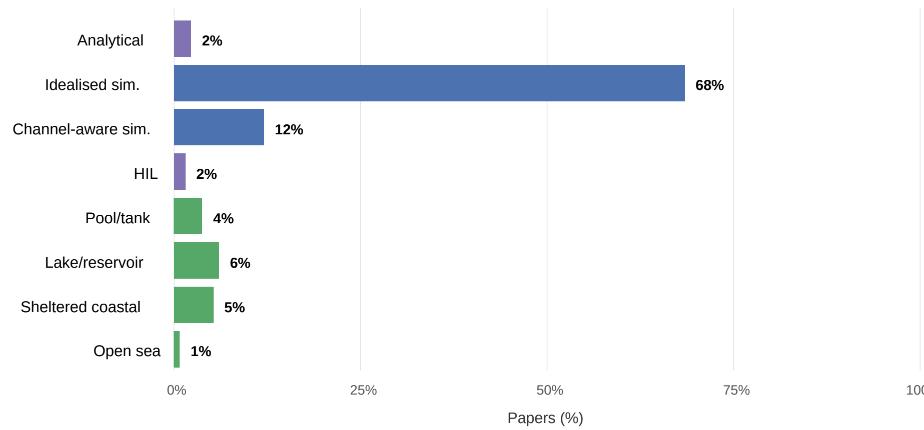

Fig. 6. Validation evidence rung for the 133 primary studies. The corpus is dominated by idealised simulation; bars show percentages of primary studies. Channel-aware simulation and physical validation are much less frequent, and open-sea evidence is rare.

indeterminate on actuation, because many papers at least state a vehicle model or reference class. The asymmetry is substantive. The corpus usually describes what the coordination law computes, but less often describes the sensing, localisation, channel-access and energy conditions under which that computation remains executable underwater.

The validation ladder in Figure 6 sharpens the scope of the contract evidence. Sixty-eight percent of studies rely on idealised simulation, 12% on channel-aware simulation and only 17% provide pool/tank, lake/reservoir, sheltered-coastal, open-sea or hardware-in-the-loop (HIL) evidence. Simulation remains useful, but it supports only the envelope it models. Pool tests can validate software integration and local dynamics but not kilometre-scale propagation or fleet-level contention; channel-aware simulations can test acoustic assumptions while leaving actuation, sensing and energy idealised. The family analyses below therefore ask which contract or envelope dimensions make each family's evidence category come out as it does, rather than treating the paper-level label as a verdict on the intrinsic value of the method.

6.1 Formation and Leader-Follower Control

Formation and leader-follower control is the largest family in the corpus, with 57 primary studies, and the one whose $\mathcal{IC}$ is most tightly coupled. A formation exists only if a leader or neighbour state is communicated (C1), arrives fresh enough to act on (C2), is placed in a common frame through relative localisation (C3), and is converted into a reference the vehicle can physically execute (C4). These four constraints must hold together, in the same study, for the claim to be auditable. Figure 7 shows why this coupling is decisive. Read constraint by constraint, the family looks well specified: communication is declared in 65% of the studies and actuation in 72%. Read as a coupled contract, the picture inverts. The share of studies that close the whole chain jointly falls from 65% at communication to 46% once freshness is added, to 11% once localisation is required, and to 9% once an executable reference is demanded. Only 9% of studies close the coupled C1–C4 loop, and requiring observability as well leaves only the digital-twin field study as a conditionally compatible case. The collapse is not gradual; it happens at localisation. Before that collapse, the family divides into two broad strands. A first strand advances the controller while leaving the underwater communication and localisation substrate mostly implicit. It gathers intermittent, event-triggered and predictive designs [52, 89, 107, 128, 171], fixed-time and predefined-time schemes [44, 72, 123, 160], sliding-mode and neuro-adaptive controllers [32, 88, 120, 126],

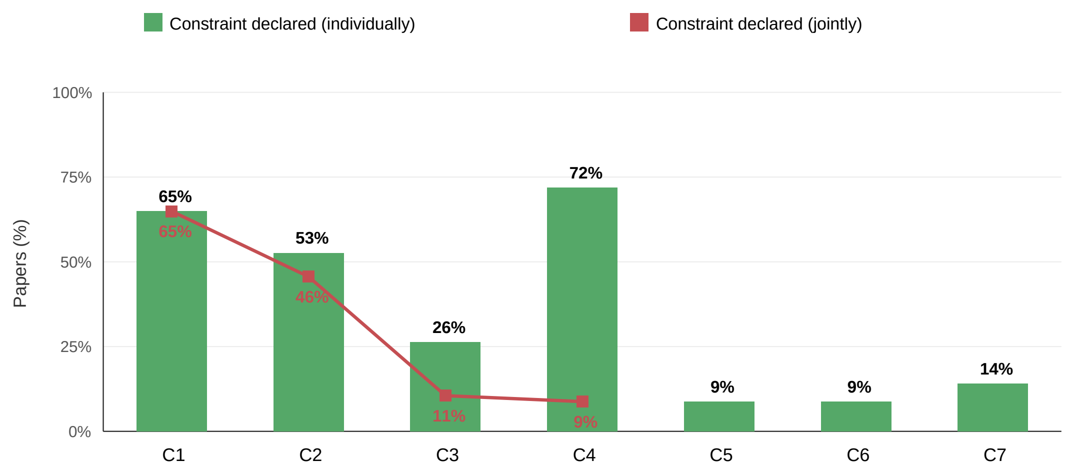

Fig. 7. Green bars give the share that declares each constraint in isolation (Demonstrated or Conditionally Compatible). The red line gives cumulative joint closure along the coupling chain: the share that closes every constraint up to that point in the *same* study. Individually the chain constraints C1–C4 look 26 to 72% covered, but joint closure falls from 65% at C1 to 9% at C1 through C4, breaking at localisation (C3); adding observability (C5) leaves only the 2% digital-twin field case. Separately, 25% of studies command a reference the declared platform is not shown to deliver, the source of every potentially incompatible verdict in the family.

discrete-time trajectory-tracking formulations [65, 152, 158], leader-follower, path-following and three-dimensional variants for underactuated vehicles [5, 9, 30, 45, 149], and publish-subscribe or automaton-based coordination [36, 143]. These designs typically improve convergence, robustness or fault tolerance, but treat the arrival of neighbour state as given. A second strand instead moves parts of the acoustic contract into the formation design itself: multi-rate and surfacing or event-based updates [96, 111], dynamic and asynchronous event-triggered exchange [12, 119, 153], explicit models of delay, packet loss, dropout and prediction [13, 63, 69, 82, 108], and formation costs that respond to signal-to-noise ratio or path loss [16, 55]. Weak-communication, long-delay and reconfiguration studies extend the same tendency [90, 115, 138, 156, 169]. The intra-family pattern is therefore not an absence of communication-aware work; it is that communication and freshness are frequently closed before the rest of the contract is. The family also carries most of the corpus's potentially incompatible actuation evidence, 14 of its 20 cases: 25% of studies command a reference the declared platform is not shown to deliver. These span target-hunting, neural and cooperative-control formulations [17, 33, 93], fixed-time, sliding-mode and decoupled controllers [44, 72, 123, 155], event-triggered, surfacing and leaderless variants [68, 111, 124, 150], and path-parameter, heterogeneous and bipartite schemes [20, 125, 129]. The issue is narrow rather than generic: the geometry is proved over a vehicle abstraction, often fully actuated or reduced in degrees of freedom, more capable than the declared underwater platform substantiates. Most of the family remains demonstrated or conditionally compatible on actuation, so this sub-strand is the exception that produces every incompatible verdict rather than a statement that formation control is physically unrealistic. Localisation is where the coupled contract breaks. Relative localisation is left indeterminate in three quarters of the family, even though maintaining a geometric relationship is the definition of a formation. The minority that closes it does so by declaring a positioning mechanism as part of the design, whether acoustic ranging, transformed relative states, adaptive or velocity-free observers, or explicit positioning infrastructure [8, 37, 45, 115, 120, 162]. These are also the studies that approach a complete contract, which is consistent with the joint-closure curve: localisation, not communication, is the constraint that decides whether the rest of the chain can matter. Physical and channel-aware evidence strengthens individual claims without removing the pattern. Channel-aware simulations expose path loss, signal-to-noise ratio, packet loss or communication-energy cost inside the loop [8, 13, 16, 55, 119]; pool and tank trials surface integration effects that idealised simulation misses [108, 124]; and lake or reservoir experiments make delay, dropout, modem use and finite resynchronisation windows concrete [12, 69, 116, 162]. Sheltered-coastal evidence adds the strongest system pressure, particularly where acoustic modems, surface support or localisation assets are part of the trial [56, 94, 142]. The digital-twin field study is one of the family's strongest real-hardware cases, combining underwater acoustic communication, dual positioning infrastructure and energy-aware networking across pool, lake and near-shore validation [142]; it is the family's conditionally compatible case because the integrated stack functionally supplies the timing needed for twin matching, while the clock or timestamping mechanism itself is not characterised. The system-level simulation that comes closest declares buoy localisation, communication relays, signal-to-noise-aware formation cost and per-bit energy as one envelope [16], yet remains indeterminate because its long-range relay motivation is never validated at the claimed scale.

The lesson is precise. Formation evidence is strongest when geometry, acoustic update, localisation support, sensing footprint, actuation envelope, energy horizon and channel access are declared as a single system. When only the controller is specified, the outcome is a coherent formation law rather than an auditable underwater coordination claim, and the coupled contract that defines the family stays open.

6.2 Consensus and Distributed Control

Consensus and distributed control is the corpus's most mathematically developed family and its least operationally grounded. Its 23 studies reason about agreement over a communication graph, treating connectivity, switching topology and communication delay as explicit objects and proving convergence by construction. The Information Contract shows that these proofs specify what they contain and leave unstated what they presuppose. The three quantities a networked-control formulation writes down, the vehicle model (C4), the delay it tolerates (C2), and some account of the channel that carries the graph edges (C1), are declared by 70%, 39% and 35% of the studies. The four that a convergence proof simply assumes, localisation (C3), sensing and observability (C5), energy (C6), and medium access as the fleet scales (C7), fall to 9%, 0%, 0% and 4%. Taken across these four constraints together, consensus specifies its deployment substrate less than any other family in the corpus (Figure 8). The control-theoretic core is genuine, and it is where the family invests. Leader-following and distributed formation-consensus laws establish agreement over discrete-time and switching topologies with formal convergence guarantees [48, 64, 151, 154, 157], extended by average-consensus and time-varying formation tracking [144, 148, 161] and by coordinated path following [46]. A second line treats the graph itself as the object of study through static-consensus conditions and distributed synchronisation [77, 147], while observer-based, event-triggered and fixed-time designs reduce how much and how often state must be exchanged [81, 140, 163]. In the tradition of classical consensus theory, which treats switching topology and communication delay as explicit mathematical objects [83], delay and topology are modelled here with real care, from bounded time-varying delays to Markov-switching topologies and LMI stability conditions. These are the terms the theorems need, and they are supplied. The deployment substrate is where the family stops, and the stop is abrupt. Localisation is declared in 9% of studies, medium access in 4%, and sensing range and energy in 0%. Most papers assume each vehicle knows its own and its neighbours' states exactly, with no model of inertial or Doppler drift, acoustic ranging error, or cooperative positioning. This is not a reticence consensus merely shares with the field: the rest of the corpus declares these same constraints

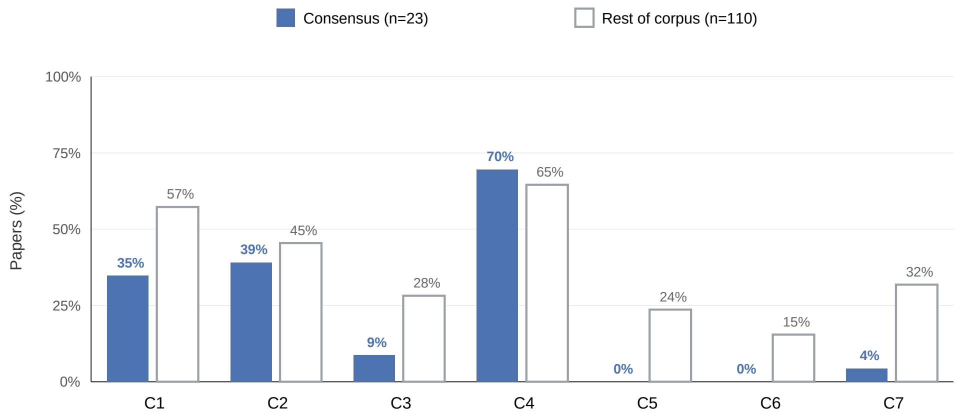

Fig. 8. The consensus family against the deployment substrate. For each constraint, bars give the share of the 23 consensus studies (filled) and of the other 110 corpus studies (outlined) that declare it (Demonstrated or Conditionally Compatible). Consensus specifies the constraints a networked-control proof contains, the vehicle model (C4), delay (C2) and channel (C1), but the constraints it presupposes, localisation (C3), medium access (C7), sensing (C5) and energy (C6), fall to 9%, 4%, 0% and 0%, at or near the corpus floor. Sensing and energy are declared by no consensus study, and only 4% are validated in water.

several times as often, and consensus sits at or near the corpus floor on all four. The pattern has one revealing exception. The 4% of the family evaluated in water, a mixed pool experiment with two real vehicles alongside simulated ones, is also the only case to demonstrate localisation, using dead reckoning with occasional acoustic-beacon fixes and an ID-ordered token-passing channel, because operating real hardware forces a positioning and medium-access solution the simulations never have to name [58]. The closest the rest of the literature comes to confronting the channel is analytical: one topology study notes that its consensus-optimal complete digraph would be prohibitively expensive for low-power AUVs on an acoustic link and retreats to a spanning tree, an observation it never validates [137]. Every other consensus study runs in idealised simulation, so the graph is analysed but never carried on a physical acoustic network. The omission has a measurable cost: consensus carries the highest potential-incompatibility rate in the corpus, 30% of studies. Most of these cases prove agreement over a vehicle more capable than a real underactuated AUV, commanding a direct three-axis force that includes sway and treating the platform as a fully actuated point mass with no thrust allocation; the pattern recurs across finite-time, observer-based, neurodynamic and learning-based designs [19, 21, 29, 106, 145, 166]. The remaining case fails on the channel rather than the hull: it fixes a one-second communication delay while claiming a 2500-metre coverage range whose acoustic propagation floor alone is 1.67 seconds, so the assumed exchange conflicts with acoustic propagation at the stated scale [38]. In every case the agreement is provable and the operating precondition that would let it run is absent.

The lesson is narrow and exact. Consensus has developed a mature mathematics of agreement over a graph. What it has not stated, except in the 4% of studies taken into the water, is where the vehicles are, what they sense, what they cost to run, and how they share the channel, the physical preconditions on which the graph assumed by its proofs depends.

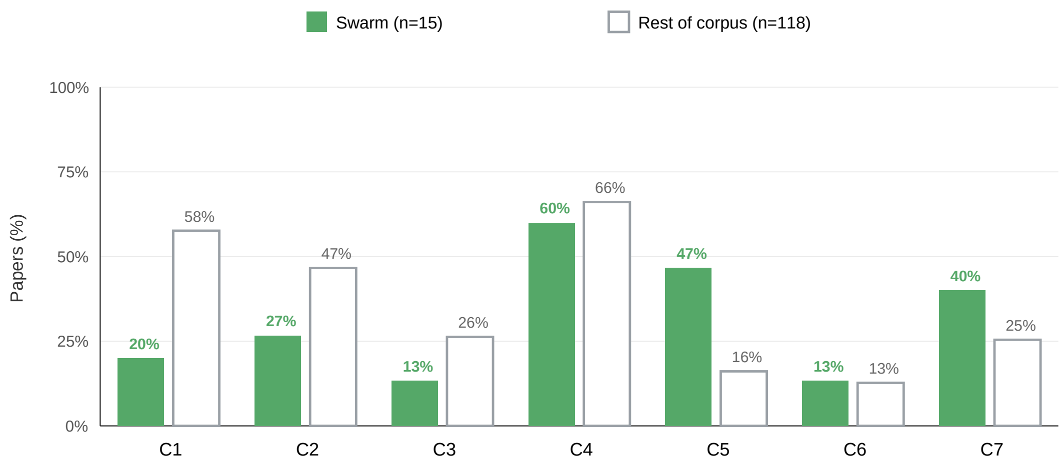

Fig. 9. The swarm inversion. For each constraint, bars give the share of the 15 swarm studies (filled) and of the other 118 corpus studies (outlined) that declare it (Demonstrated or Conditionally Compatible). Swarm is the corpus minimum on the acoustic channel (C1, 20%) and the corpus maximum on local sensing (C5, 47%), the only family in which sensing is specified more often than the channel, with lightweight local medium access (C7) following the same logic. Absolute heights are retained: only actuation (C4) clears half, so the inversion is a lean toward local perception, not a grounding of it.

6.3 Swarm and Bio-Inspired Methods

Swarm and bio-inspired methods invert the coordination substrate. Where formation and consensus primarily route coordination through communicated state, these 15 studies shift the burden toward local perception, the principle by which natural collectives coordinate through local sensing rather than global positioning [92]. Swarm is the corpus minimum on communication (C1, declared by 20% of its studies against 58% elsewhere) and the corpus maximum on observability and sensing (C5, 47% against 16%), the only family in which sensing is specified more often than the channel (Figure 9). The reading has two layers. Relative to the corpus the inversion is sharp: swarm sits below every other family on the channel and above every other family on sensing, with lightweight local medium access (C7, 40%) following the same local-first logic. In absolute terms it is modest: only actuation (C4, 60%) clears half, and even sensing, the primitive the family leans on, is characterised in fewer than half its studies. The inversion changes which constraints swarm engages, not whether it grounds them. In the strongest strand, the channel is divested by design rather than merely omitted. A minimal-signalling strand reduces inter-vehicle communication until the acoustic constraint stops binding: bio-inspired TDMA arbitrates the medium through local slot adaptation rather than central scheduling [109], pheromone-inspired monitoring and dual-trail stigmergic coordination replace messages with virtual environmental fields that each vehicle deposits and reads [62, 141], and distributed shape control drives the collective from local relative geometry alone [59]. These designs are why swarm sits at the corpus floor on communication: coordination is carried by the environment and by local rules, so the channel is asked to do very little. That same move quietly relocates the difficulty into sensing, and here the family divides almost evenly. In 47% of studies, local perception is at least declared or bounded: a range and neighbour set in Lagrangian particle-swarm control [102], forward sensing for bio-inspired obstacle avoidance [127], the onboard perception of the self-aware CoCoRo swarm [100], a one-bit detector for collective event response [113], and declared sensing assumptions in a differential-game learner, a division-of-labour swarm and an adaptive navigator [74, 131, 159]. In the remaining studies it is virtual. Target and neighbour states are assumed available without a sensor in cooperative tracking, hybrid coordination, coverage planning and consistent collaboration [11, 98, 122, 136], several of them learning-based policies trained in simulators whose faithful acoustic modelling is only now emerging [47]; or the sensed quantity is an idealised field, a pheromone or a stigmergic trail, rather than a measured acoustic or optical return [62, 141]. Much of what the family reports as sensing is therefore an abstraction: swarm leans on perception more than any family, yet grounds it in a declared sensing mechanism or range in only 47% of studies, and in a fully physical underwater sensing model in fewer still. This is not a simulation artefact. Two of the family's three in-water trials, bio-inspired TDMA and pheromone monitoring, still leave sensing indeterminate [62, 109], because the abstraction is a modelling choice inherited from terrestrial swarm robotics, not a consequence of staying dry. It is the swarm counterpart of the graph-as-algebra gap in consensus: the coordination primitive is assumed rather than measured. The inversion has one clear benefit: no swarm study is coded as potentially incompatible under our framework. It reaches that position from the opposite direction to the architecture family, which declares its infrastructure. Swarm instead demands very little: it does not over-claim the channel it barely uses, and its reactive local control is specified without the fully-actuated idealisation that produces incompatibility in formation and consensus. But absence of detected conflict is not closure. The constraints the inversion leaves untouched are the corpus-wide ones, localisation and energy, each declared in only 13% of studies, so every swarm study remains indeterminate overall, none demonstrated or conditionally compatible.

The lesson is that minimising communication does not remove the underwater contract, it moves it. Swarm and bio-inspired methods relieve the acoustic channel, the constraint that stresses formation and consensus, but they pay for it with a sensing and localisation burden they mostly assume rather than measure. The family is the corpus's clearest evidence that coordination can be designed around the channel, and its clearest reminder that doing so still requires the vehicle to perceive, locate and power itself, which these studies have yet to specify.

6.4 Task Allocation and Mission Planning

Task allocation and mission planning is the assignment layer of the corpus. Its 18 studies decide who does what, through auctions, markets, contract nets and learning-based schedulers, and their constraint profile splits along exactly that role. Measured against the rest of the corpus, the family sits below on the whole execution substrate and above on the terms of the assignment itself (Figure 10). It is below the corpus on the acoustic link that carries the bids (communication C1, 44% against 55%; freshness C2, 28% against 47%) and on the vehicle that carries out the task (localisation C3, 6%; actuation C4, 44% against 69%), and above the corpus on what the assignment optimises (task observability and sensing support C5, 44% against 16%; energy C6, 28% against 10%; scale C7, 44% against 24%). The split is clean: every execution constraint falls below the corpus, every objective constraint above it. As elsewhere the reading has two layers. Relative to the corpus the pattern is sharp. In absolute terms it is modest: none of the constraints task allocation leads on clears 44%, so the family specifies its objective more than its execution and closes neither, with 83% of studies indeterminate, 11% potentially incompatible and 6% conditionally compatible.

What the family specifies is the assignment and its cost. The mechanisms are well developed: market and auction frameworks price tasks and let vehicles bid [41, 168], contract-network and hierarchical schemes negotiate assignments [66, 91], and reinforcement-learning, self-organising-map, repulsion and search-and-rescue allocators optimise the mapping under various objectives [10, 67, 73, 75, 134, 173]. Energy is the one constraint on which task allocation genuinely leads rather than merely sits above the corpus average: it holds the most demonstrated energy evidence of any family, because energy often enters the assignment cost directly. Distributed mission planning, heterogeneous hunting

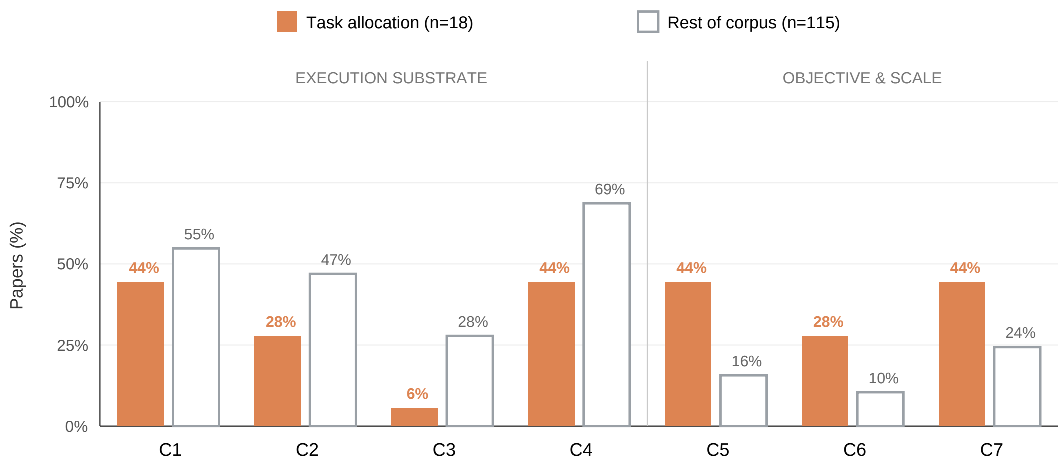

Fig. 10. Task allocation, objective versus execution. For each constraint, bars give the share of the 18 task-allocation studies (filled) and of the other 115 corpus studies (outlined) that declare it (Demonstrated or Conditionally Compatible). The profile splits cleanly: task allocation is below the corpus on the execution substrate, the link (C1, C2) and the vehicle (C3, C4), and above it on what the assignment optimises, task observability and sensing support (C5), energy (C6) and scale (C7). Absolute heights are retained: no constraint it leads on clears 44%, so the lean toward the objective is relative, not a grounding of it.

allocation and networked assignment carry validated energy or endurance budgets as the assignment cost [60, 78, 164], and auction and static-search formulations declare consumption models alongside them [71, 168]. Coverage and scale follow the same logic: surveillance-network and mine-countermeasure missions define the tasks by area and target and analyse how the assignment scales with the team [40, 99]. What the family assumes is the execution. Localisation is specified in exactly 6% of studies [60]; the remainder take each vehicle's position as a known input to the assignment cost, with no model of drift or ranging error. Actuation is the corpus minimum, sixteen points below the next family: only a handful state a vehicle model at all, mostly the learning-based and hunting allocators that need one to simulate the plan [75, 164, 173], while the rest treat the assigned task as executable by construction. As a result task allocation is the only family with no physical validation whatsoever, 100% of studies are simulation, so the assumption that a vehicle can locate itself and reach its assigned task is never tested against a hull. The family's potential incompatibilities show what confronting that assumption costs. Both are the channel-aware studies that stopped assuming the link and modelled it. One parameterises range-dependent packet loss and propagation delay and finds a connectivity tipping point beyond which consensus-based bidding fragments and the assignment fails to converge [118]; the other couples allocation with path planning under frame-error and network limits and reaches a comparable infeasibility boundary [34]. The contradictions in this family come precisely from the studies honest enough to model the communication substrate the rest take for granted.

The lesson is that task allocation has built the plan without the means to run it. Its mission logic is well developed and its energy accounting is comparatively strong, yet localisation, actuation and the acoustic link, the substrate any assignment must execute over, are exactly the constraints it leaves to be assumed. It is the corpus's recurring shape once more: each family closes the layer it works in and abstracts the one beneath, and here the abstracted layer is the execution.

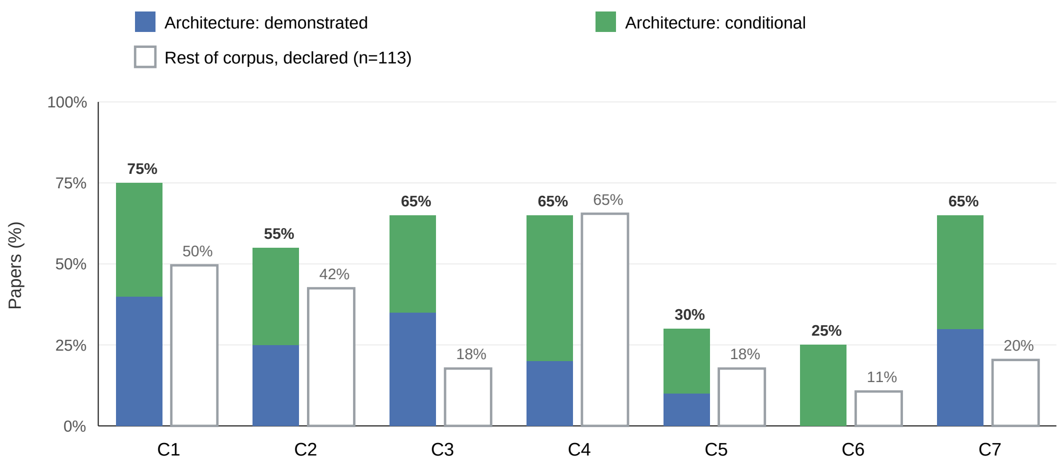

Fig. 11. Architecture as the supply side. For each constraint, the filled bar gives the share of the 20 architecture studies that declare it, split into Demonstrated and Conditionally Compatible, against the outlined share of the other 113 corpus studies. Architecture is the corpus maximum on the communication and localisation infrastructure the other families assume, the channel (C1), freshness (C2), localisation (C3) and medium access (C7), and its lead on C3 and C7 is the largest of any family on any constraint. Every constraint carries a demonstrated base except energy (C6), which is declared conditionally in 25% of studies and demonstrated in 0%.

6.5 Architecture and Support Systems

Architecture and support systems is the supply side of the coordination contract. Its 20 studies are not coordination methods in the same sense as formation, consensus, swarm or task allocation; they supply the infrastructure those methods presuppose: cooperative localisation and navigation systems, communication stacks, relay and beacon designs, and mission-management platforms. Against the rest of the corpus its profile is a supply-side benchmark: architecture/support papers are included because they make explicit the infrastructure the other families often leave implicit. It is therefore the maximum on precisely the constraints the others assume: the acoustic channel (C1, 75% against 50%), freshness (C2, 55% against 42%), localisation (C3, 65% against 18%) and medium access and scale (C7, 65% against 20%), and its lead on localisation and medium access, forty-seven and forty-five points, is the largest above-corpus gap of any family on any constraint. It does not merely declare this infrastructure, it fields it: at family level, architecture contains demonstrated evidence on six of the seven constraints and is the most field-validated family in the corpus, with 40% of studies in sheltered-coastal, lake or open-sea trials (Figure 11). That is why it holds three of the corpus's five conditionally compatible studies. The single exception, the one bar in the figure without a demonstrated base, is energy: C6 is declared in 25% of studies and demonstrated in 0%.

What the family supplies, and measures, is localisation and communication. Its strongest evidence is a run of cooperative-navigation systems that put relative positioning in the water: origin-state and single-beacon estimators for communication-constrained fleets [114, 130], a single maneuvering surface craft used as a moving beacon [39], synchronous-clock range-and-angle relative acoustic navigation [97], and the MORPH communications and relative-navigation architecture trialled in a harbour [43], with cooperative-SLAM formulations folding mapping into the estimate [84, 133]. The communication side is as explicit: coordination architectures for UUV fleets and multi-agent efficiency [42, 103], hybrid-automaton and gesture-based coordination [23, 135], and relay-scheduling designs that maintain coordination under declared channel limits [80, 139]. Even the intervention subgroup, cooperative manipulation and impedance control for vehicle-manipulator systems, is coded here for the coordination and sensing architecture it specifies [28, 53]. The modems, schedules, ranging beacons and relays the other four families assume are here named, parameterised, and in the field studies measured, from EvoLogics and WHOI modem rates to TDMA and one-way beacon cadences. That grounding is why architecture closes more of the contract than any family, and its conditionally compatible studies are its most complete. The GREX sea trials declare a DSPComm AquaComm 480 bps link, 19-byte packets on a 12 s TDMA schedule, and platform endurance figures, and validate two real AUVs in sheltered coastal water [6]. The Xiroi II surface platform raises acoustic positioning from 0.118 to 0.468 Hz with a declared 1480 Wh battery and real relay validation [79]. The cooperative-localisation study quantifies roughly 100 bytes per second of throughput, 32-byte packets and 10 to 30 s schedules with real WHOI modems across multiple coastal trials [4]. Each names a communication rate, a physical positioning infrastructure and an energy figure, and validates on hardware. The rest fall short of closure on one dimension or another, whether an uncharacterised channel in a hardware-in-the-loop study [50] or, most often, an absent energy budget; observability is the quieter soft spot, indeterminate in 70% of studies because environmental sensing is delegated to the application layer. Energy is the exception. Architecture has demonstrated evidence on the channel (40%), localisation (35%), medium access (30%), freshness (25%), actuation (20%) and observability (10%), and on energy, 0%. The 25% of studies that address it, including the one built around an energy-aware coordination protocol with an explicit per-bit transmission cost [70], declare a budget conditionally but never validate that coordination is sustainable within it; even the conditional cases offer platform battery specifications rather than a mission-level energy budget. The family that fields the infrastructure to make coordination feasible still does not demonstrate the energy budget that would keep it running.

The lesson is that architecture and support systems is the family that comes closest to closing the $\mathcal{IC}(m \mid \Omega)$ this survey proposes, and the clearest evidence that contract closure must be treated as a system property: where the communication, freshness, localisation and medium-access substrate is declared and validated, coordination moves from asserted to auditable, and the conditionally compatible studies do so on that basis. But it does not close the contract. No study reaches demonstrated compatibility, because the one constraint even the supply-side family does not demonstrate is energy, the same constraint that stops every other family. Architecture supplies what the algorithms assume; energy is what architecture, in turn, leaves open. Closing the contract underwater will require the family that has fielded the channel and the beacon to also demonstrate the budget over which they run.

7 Implications for Future Research

The $\mathcal{IC}$ analysis of the five families points to one structural finding: the corpus holds the pieces of feasible underwater coordination but does not assemble them into a full-contract demonstration in a single study. Each algorithm family closes the layer it works in and leaves a substrate layer open: consensus leaves observability and energy unstated, swarm grounds real sensing in only 47% of studies, and task allocation specifies localisation in 6%. Architecture supplies much of that substrate, but demonstrates energy in 0% of its studies. No coded study both combines a specified coordination law with fielded infrastructure and demonstrates every relevant contract dimension, which is why 0% reach demonstrated compatibility.

We translate this into five reporting obligations and one validation obligation (Table 4). The first five make implicit contracts auditable; the sixth names the integrated trial needed for demonstrated compatibility.

Table 4. The five reporting obligations and one validation obligation implied by the $\mathcal{IC}$ analysis. The first five turn recurring evidence gaps into concrete statements a paper can make; the sixth identifies the integrated trial needed for demonstrated compatibility.

Obligation	Constraints	Evidence gap in the corpus	Minimal reporting
Coupled Contract	C1-C4, jointly	Read separately, formation declares C1 in 65% and C4 in 72%, but only 9% close the whole C1–C4 chain in one study, collapsing at localisation	Which constraints the claim closes together, over one operating scope
Freshness Contract	C2	Control update rate reported as if it were the admissible age of remote state	Maximum age of consumed state, and the communication edge over which it travels
Localisation-Energy Loop	C3, C6	Indeterminate in 75% and 87% of studies; energy demonstrated in only 4% overall and 5% of field studies	State source and drift bound; the energy budget over which the acoustic schedule is sustainable
Grounded Sensing	C3, C5	Half of swarm's sensing is a virtual field; a real sensor or range is declared in 47% of studies	A sensing range and a drift-bounded frame, not assumed or idealised perception
Validated Scale	C7	Scale claimed beyond the tested n, ignoring propagation delay, contention and localisation traffic	Evidence at the claimed fleet size, or an explicit statement that scalability is unvalidated
Algorithm on Infrastructure	All seven	Integrated trials remain partial: 0% demonstrate all contract dimensions on a fielded stack	One trial running a specified coordination law on a fielded communication and localisation stack with timing, sensing, actuation, energy and MAC characterised

The Coupled Contract. *Report the constraints a claim closes jointly, not a tally of constraints addressed separately.* Feasibility underwater is a conjunction, but the corpus reports it as a list. Formation is the clearest case: read constraint by constraint its studies look ready, declaring communication in 65% and actuation in 72%; yet the share that closes the whole communicate, keep fresh, localise and execute chain within a single study is 9%, and it collapses at localisation. A coordination claim should therefore state which constraints it closes together under one operating scope.

The Freshness Contract. *Declare the maximum age at which consumed state is still valid for the decision, and the edge over which it travels.* A controller sampling period or simulation step is not this contract; it states how often a local controller acts, not how old remote information may be when it does. Where freshness tolerance is neither declared nor derivable, the correct outcome is indeterminacy; where it is declared alongside the communication-edge length, the $\mathcal{IC}$ screen can test it against the acoustic propagation floor. The fix is one auditable sentence, "the controller tolerates communicated leader position up to X seconds old", with the edge length over which the claim holds.

The Localisation-Energy Loop. *State how vehicle state is obtained and the energy budget over which that acquisition is sustainable.* Localisation and energy are the two most frequently missing dimensions, indeterminate in 75% and 87% of studies, and they are coupled: acoustic ranging and cooperative localisation bound drift but consume channel time and power, while reducing acoustic exchange saves energy but worsens drift and information age. Energy is the residual gap: entering the water lifts localisation declaration from 14% to 81% and communication from 47% to 86%, but energy only from 12% to 19%. It is demonstrated in 4% of all studies and 5% of field studies, and in 0% of architecture studies. The obligation is to turn platform endurance, battery and modem figures [4, 6, 79] into a mission-level budget tied to the coordination schedule.

Grounded Sensing. *When perception replaces communication, declare a real sensor and a drift-bounded frame, not a virtual field.* Swarm and bio-inspired methods reduce channel burden, but half of their sensing is abstract: only 47% declare a real sensing mechanism or range, while the rest use assumed neighbour states or idealised fields [62, 141, 159]. Replacing communication with perception relocates the feasibility question to spatial correspondence, which is a localisation and observability problem.

Validated Scale. *Provide evidence at the fleet size claimed, or state that scalability is unvalidated.* The underwater effects of a larger fleet are not only a larger n in a graph: traffic load, propagation delay, collision-avoidance windows, scheduling overhead, localisation traffic and packet loss all change with fleet size and geometry, so a protocol validated at $n = 5$ does not by itself establish behaviour at $n = 20$. The bio-inspired TDMA study is the template: packet loss is quantified as a function of fleet size from 10 to 80 robots in simulation and cross-checked on four real robots [109]. Formation and consensus methods should show that their per-vehicle communication and localisation burden stays sustainable at scale; task-allocation methods should compare negotiation load with realistic acoustic capacity.

Algorithm on Infrastructure. *Run a specified coordination law on a fielded communication and localisation stack, reported as one system.* This is the missing demonstrated join, and the reason 0% of studies reach demonstrated compatibility. The two halves already exist: the architecture family has fielded the modems, schedules, ranging beacons and relays, with demonstrated communication in 40% of its studies, localisation in 35% and medium access in 30%, while the algorithm families hold the formation, consensus, swarm and allocation laws. The strongest integrated algorithm case is conditional rather than demonstrated because its timing basis is characterised only functionally. The next step is to assemble the infrastructure and algorithm with every contract dimension declared and evidenced in the same trial.

Together the six obligations reduce to one requirement: report the operating envelope Ω alongside the coordination law, including edge length, fleet size, topology, localisation, channel/MAC, synchronisation, mission horizon, energy, infrastructure and environment. These are not new experiments; they are the quantities the strongest studies already report and less complete studies leave implicit. The agenda is not to prefer one family over another, but to make clear which constraints a scheme closes by design, which it resolves through declared infrastructure, and which remain validation obligations.

8 Threats to Validity

This paper is a secondary study: its findings are inferences drawn from primary literature through an explicitly defined apparatus: the $\mathcal{IC}$ model, the operating-envelope reconstruction, and the per-study constraint coding, rather than direct measurements of deployed systems. Its conclusions are therefore only as sound as that apparatus and the corpus to which it was applied. We assess threats following standard validity categories for secondary studies [2, 132].

Construct Validity concerns whether the measurement apparatus captures the property it is meant to capture. The governing threat is framework relativity: a study coded as indeterminate or potentially incompatible has not necessarily failed on its own terms, but has been re-evaluated against a feasibility definition it did not adopt. If the $\mathcal{IC}$ model mis-specifies what underwater coordination requires, the evidence labels inherit that mis-specification. We make this dependence explicit throughout: the findings are consequences of the $\mathcal{IC}$ framework applied to the corpus, not theory-neutral observations. We mitigate the threat by deriving the seven constraints from recurring assumptions and limitations in the corpus (Section 5), and by grounding C1–C7 in physical or operational quantities such as propagation speed, drift, sensing range, actuation limits, energy use and channel access.

Two operationalisations carry particular construct risk. First, C2 depends on the maximum admissible information age required by a coordination mechanism. A controller sampling period or simulation step is not, by itself, such a contract: it states how often a local controller computes an action, not how old remote information may be when that action is consumed. Accordingly, when no delay tolerance is declared and no mechanism-specific derivation is possible, C2 is coded indeterminate. A potential freshness mismatch is assigned only when both the relevant freshness requirement and the communication-edge length are reported or defensibly derivable. Second, $\tau_{\min} = r/c$ depends on the relevant edge length r, which is sometimes inferred rather than stated. We take r to be the communication or sensing edge length, for example leader–follower separation or neighbour distance, not the total mission area or vehicle travel distance. When r is inferred from simulation geometry or mission description, the inference rule is recorded in the coding table. Thus, potential freshness mismatches should be interpreted as framework-relative evidence about declared or derivable contracts, not as proof that a mechanism is physically impossible.

A further construct risk concerns the scope of C7. Medium-access and scalability evidence is meaningful only relative to the fleet size, geometry, traffic model and MAC/channel assumptions claimed by the study. The risk is two-sided: abstract simulations may overstate scalability if they omit propagation-delay contention, localisation traffic or packet loss, while small-team experiments should not be penalised when the paper claims only that tested scale. We therefore code C7 against the declared operating envelope: evidence supports larger-scale claims only when the evaluation or analysis represents the corresponding fleet size, channel-access regime and traffic load; otherwise the evidence remains scope-limited or indeterminate.

A final construct threat concerns family assignment. Sorting each study into one of five families can flatten hybrid mechanisms. We treat families as analytic lenses rather than mutually exclusive theoretical categories, and code multi-mechanism papers by their dominant primitive: the mechanism that carries the paper's main contribution and evaluation. The Architecture/support family is not a coordination method in the same sense as formation, consensus, swarm or task allocation; it is retained because it supplies the infrastructure layer that the other families often presuppose. Its rankings are therefore supply-side benchmarks, not evidence of intrinsic superiority over algorithmic families.

Internal Validity concerns whether classifications follow soundly from the evidence. The principal threat is interpretive latitude: constraint evidence, edge-length inference, infrastructure dependence and evidence category each involve judgement. Each family-coded paper was independently coded by two coders using the same extraction form. After independent extraction, disagreements were resolved by returning to the primary text and retaining a final value only when consensus was reached. The harmonised table is therefore an adjudicated consensus record rather than a single-coder judgement. We release the independent coder files, the harmonised table and the audit material so that readers can inspect both the pre-reconciliation evidence and the final decisions. One derived quantity compounds this latitude: the joint-closure count in Section 6.1 treats a study as closing the coupled contract only when every constituent constraint is independently coded as specified, so a single-constraint change can move a study across that threshold. We therefore read joint-closure figures as indicative of the coupling structure rather than as exact rates.

Publication-level counting creates a second internal-validity threat. The reported counts treat papers as units of evidence, but some papers are incremental variants from the same author group, controller family, simulator or platform lineage. Such lineages may overweight particular reporting habits or modelling assumptions. We therefore interpret counts as descriptive evidence over the indexed publication corpus, not as independent statistical samples from the population of possible coordination methods. This limitation is less damaging to the revised analysis than to an evidence-rate claim, because the central result is not a precise estimate of physical failure frequency but the broad pattern that key elements of the operating envelope are often undeclared.

We also note that the evidence taxonomy is designed to distinguish missing evidence, conditional support and potential incompatibility. A study assigned to the "potentially incompatible" category is not declared physically impossible; it is coded as having reported or defensibly derived quantities that conflict with at least one constraint under the stated envelope. A study lacking the information needed to make that determination is coded indeterminate.

External Validity concerns generalisation beyond the corpus. The dominant threat is corpus representativeness, especially the conclusion that the indexed literature is dominated by simulation and that field validation is proportionally highest in architecture and support-system studies. This pattern is consistent with prior underwater multi-robot surveys, which note the cost and logistical difficulty of field campaigns [172], and with recent simulator-focused analyses that report a persistent gap between simulation validation and real-world underwater deployment [35]. Nevertheless, open-water and defence-oriented field campaigns may be reported in venues whose indexing is uneven. We bound but cannot eliminate this threat by combining databases with complementary coverage, using recall-oriented queries, and applying backward citation chasing. The simulation-dominance finding should therefore be read as a property of the indexed peer-reviewed literature, not as a census of all multi-AUV coordination ever deployed.

A second external-validity threat concerns mixed surface–submerged systems. Studies involving surface relays, GPS-supported surface assets or centralised mission managers are included when they address submerged coordination under acoustic constraints, but their feasibility claims do not transfer automatically to fully submerged deployments. We mitigate this by recording declared infrastructure as part of the operating envelope and by treating such support as a conditional scope restriction rather than as an algorithm-only property.

A third external-validity threat concerns the implications in Section 7. The proposed research obligations are derived from the evidence patterns in the corpus and checked retrospectively against the reviewed studies. They have not been validated prospectively on a new deployment. Their status is therefore a structured reporting and evaluation workflow, not an empirically validated design methodology.

Conclusion Validity concerns whether the quantitative results are reliable and reproducible. The first threat is false precision. Several family-level counts derive from small denominators, and all percentages are descriptive ratios over the coded corpus rather than population estimates. We therefore report counts alongside percentages and avoid treating family-level rates as inferential statistics. The same caution applies to the between-family comparisons on which Section 6 relies. Statements that a family is the corpus maximum or minimum on a constraint, or that a between-family gap is the largest observed, are descriptive orderings over the coded corpus at per-family denominators ranging from 15 to 57 studies, not significance-tested differences; a ranking or a gap of a few points can turn on a small number of papers, and should be read as the shape of the evidence rather than as a measured effect size.

The second threat is reproducibility under database non-stationarity. OpenAlex, Semantic Scholar, Google Scholar and other bibliographic services evolve continuously, so an exact re-execution may return a slightly different candidate set. We constrain this by fixing the final search date (May 2026), documenting the selection flow in PRISMA form (Figure 1), and releasing the screening master, query/source metadata, peer-review status, full-text decisions, exclusion reasons, independent coder files, harmonised coding table, PDF-audit data and analysis material as a public package [3].

A third threat concerns C5 in the Architecture/support family. We classify C5 as indeterminate when architecture papers specify the coordination backbone, namely communication, localisation, relay or scheduling, but not the front-end environmental sensing modality, range or uncertainty. This is a deliberate scope decision: such papers can demonstrate conditional support for communication or localisation infrastructure without establishing the observability contract needed by a particular coordination task. Readers should therefore interpret C5 indeterminacy in this family as a boundary of the system claim, not as a flaw in the architecture contribution.

Finally, the findings are framework-relative. The $\mathcal{IC}$ model is one explicit way to organise underwater coordination feasibility; other models could produce different categories. We do not claim that the taxonomy is the only possible one.

We claim that it is explicit, internally consistent, uniformly applied across the corpus, and sufficiently documented for readers to assess how the conclusions depend on its assumptions.

9 Conclusion

This paper asked not what multi-AUV coordination mechanisms have been proposed, but under which declared physical, infrastructural and operational conditions their feasibility claims can be assessed. The seven constraints analysed here define a coupled feasibility boundary: communication, information age, localisation, observability, actuation, energy and medium access cannot be evaluated independently in the underwater setting.

Applying the $\mathcal{IC}$ framework to 133 primary studies yields one structural result: the corpus holds the pieces of feasible coordination but does not assemble them into a full-contract demonstration, and 0% of studies provide sufficient evidence to establish full-contract compatibility across all relevant dimensions. The shortfall is structured by family, and the families fail in different places. Formation control specifies its constraints separately but closes the coupled C1–C4 chain in only 9% of its studies, breaking at localisation. Consensus proves agreement over communication graphs whose physical substrate is almost always left uninstantiated, while observability and energy remain undeclared throughout. Swarm methods trade the acoustic channel for local sensing but ground half of that sensing in a virtual field rather than a real sensor. Task allocation optimises the mission over an execution it assumes, specifying vehicle localisation in 6% of studies. Architecture and support systems most consistently declares and fields the communication and localisation infrastructure the others presuppose; it holds the clearest conditionally compatible evidence and comes closest to closing the Information Contract, and it is strong evidence that contract closure must be treated as a system property.

Two facts complete the picture. First, 0% of coded studies both combine a specified coordination law with fielded infrastructure and demonstrate every relevant contract dimension, which is why 0% reach demonstrated compatibility: the corpus already contains both the supply and the demand, but does not yet join them completely. Second, energy is the residual gap, the one constraint field validation does not lift and that even the infrastructure family, which contains demonstrated evidence on six of the seven constraints, does not demonstrate. The main conclusion is therefore not that existing algorithms are physically impossible, but that the literature specifies the coordination law more completely than the underwater system that must carry it. We offer the $\mathcal{IC}$ framework as the means to make that system claim explicit, and six obligations, the coupled contract, the freshness contract, the localisation and energy loop, grounded sensing, validated scale, and running the algorithm on fielded infrastructure, as the practical steps that turn an indeterminate coordination claim into a reproducible underwater system claim.

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